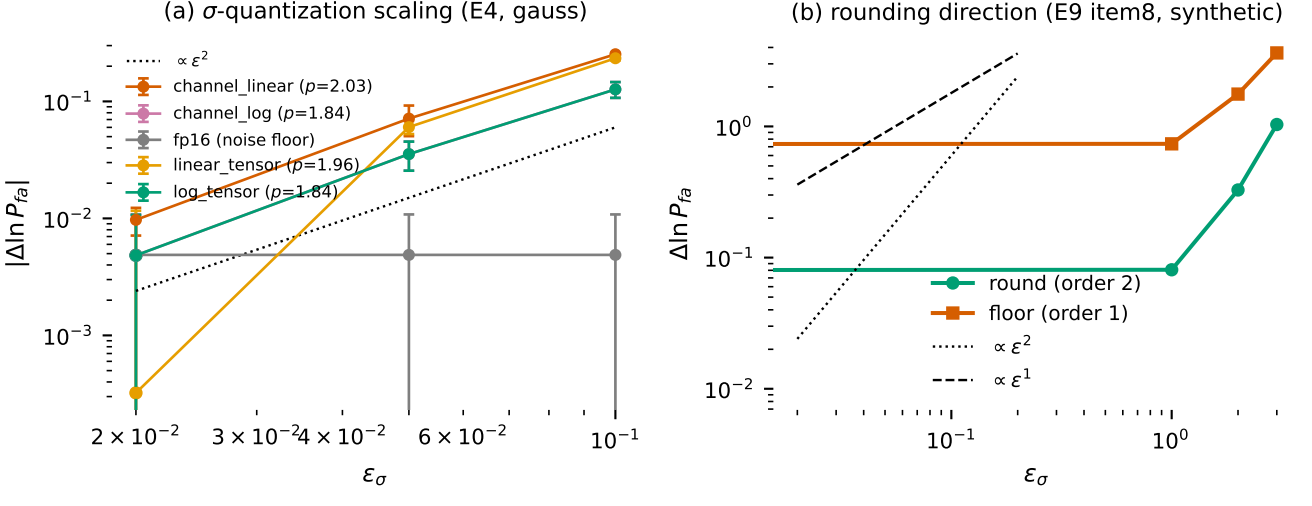
(a) σ-quantization scaling (E4, gauss)
∝ ε²
channel_linear (p=2.03)
channel_log (p=1.84)
fp16 (noise floor)
linear_tensor (p=1.96)
log_tensor (p=1.84)
|ΔlnP_fa|
10⁻¹
10⁻²
10⁻³
2 × 10⁻²
3 × 10⁻²
4 × 10⁻²
6 × 10⁻²
10⁻¹
ε_σ
(b) rounding direction (E9 item8, synthetic)
ΔlnP_fa
10⁰
10⁻¹
10⁻²
round (order 2)
floor (order 1)
∝ ε²
∝ ε¹
10⁻¹
10⁰
ε_σ